
AN0007- EC NB-IoT MQTT Application Note_V1.2

EigenCOMM Wireless Microcontroller

Document Description

This document describes the docking process between eigencomm NB-IoT chip, and cloud platform. Mqtt protocol is used for docking. This paper aims to help customers quickly complete the docking work based on eigencomm NB-IoT chip and cloud platform under the mode of at command.

Before completing the cloud platform docking in this manual, the chip (module) initialization network access process needs to be completed first, and the cloud platform business docking process can only be carried out after the module network injection is successful.

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1. Introduction

1.1 Introduction to MQTT protocol

Mqtt (message queuing telemetry transport) is an instant messaging protocol developed by IBM, which may become an important part of the Internet of things. The protocol supports all platforms, and can connect almost all networked items with the outside, which is used as the communication protocol of sensors.

Mqtt protocol is designed for communication of remote sensors and control devices with limited computing power and working in low bandwidth and unreliable networks. It has the following main characteristics:

1) Use publish / subscribe message mode to provide one to many message publishing and decouple the application.

2) Message transmission that is masked against the content of the payload.

3) Use TCP / IP to provide a network connection.

4) There are three types of messaging quality of service:

To more than once: the message release completely depends on the underlying TCP / IP network. Message loss or duplication can occur.

At least once: make sure the message arrives, but duplicate messages can occur.

Only once: make sure the message arrives once. This level can be used in the following situations, in the billing system, repeated or lost messages will lead to incorrect results.

5) Small transmission, small overhead, fixed length head is 2 bytes, protocol exchange is minimized to reduce network traffic.

1.2 Document purpose

The eigencomm NB-IoT chip mqtt application notes describe how to use mqtt, how to use eigencomm NB-IoT chip to connect user devices to the cloud, and how to realize the interaction between cloud and devices. It also introduces the access and communication of onenet and Alibaba Internet of things.

This paper introduces the use of eigencomm NB-IoT chip and cloud platform, describes the detailed at interaction process, which can help the developers of customers to complete the relevant application development as soon as possible.

1.3 Overall process of docking

The docking of eigencomm NB-IoT chip and cloud platform includes two preparations:

1) Cloud platform side - user registration and product creation

2) eigencomm NB-IoT chip module side - create client and interactive operation between client and platform

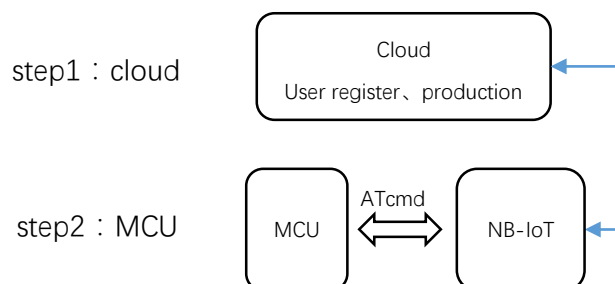


Figure 1: Docking process with server cloud

2. Docking process with OneNet platform

2.1 OneNet platform side operating instructions

2.1.1 User registration

Step 1: visit <https://open.iot.10086.cn>. Click "register" to enter the registration page

Step 2: fill in the registration information and complete the registration

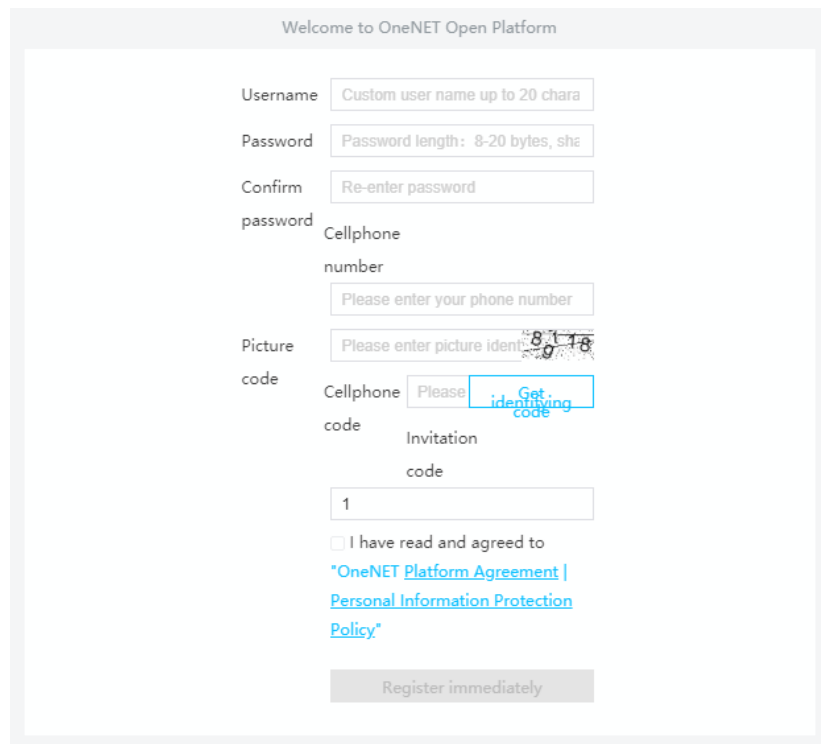


Figure 2: user registration

2.1.2 Product creation

Step 1: log in to OneNET

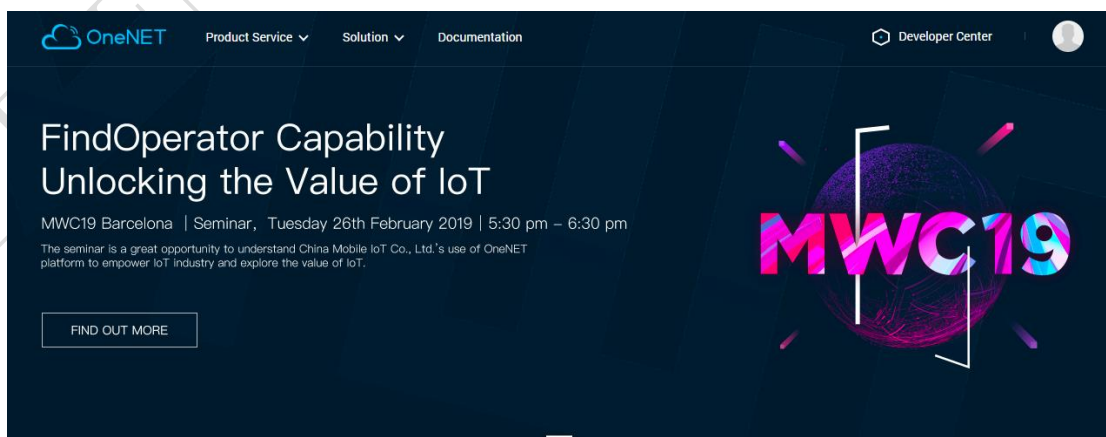


Figure 2: sign in OneNET

Step 2: click "Developer Center", enter, find the interface to enter "multi protocol access", and click to enter

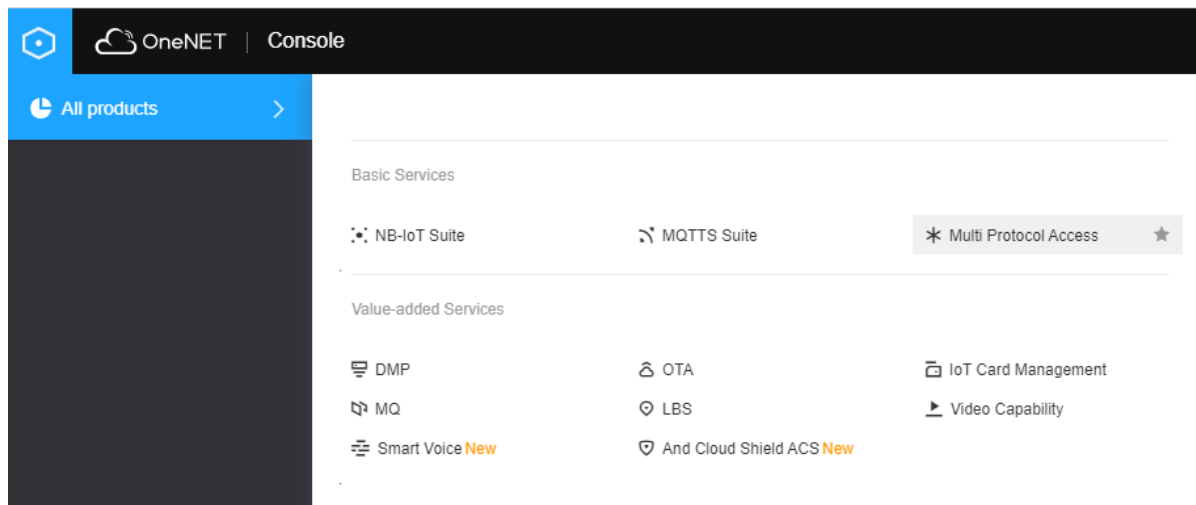


Figure 4: multi protocol access interface entry

After entering, click "add product"

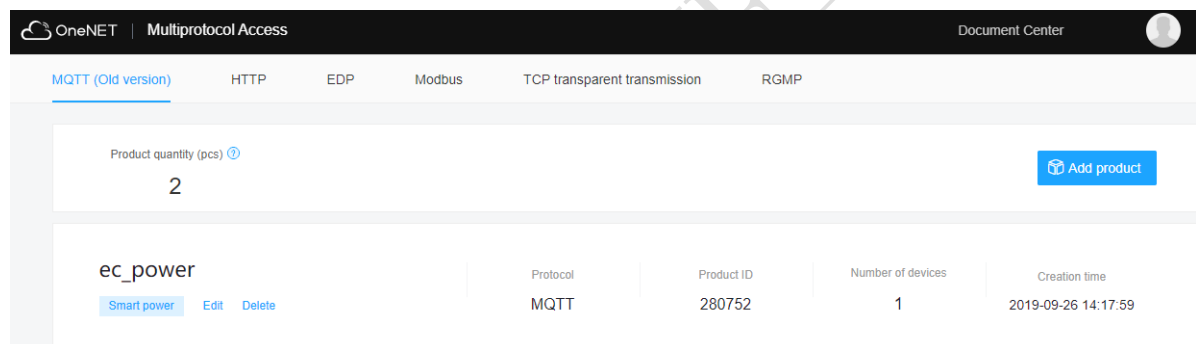


Figure 5: sign in multi protocol access interface entry

Add product

Product information

*Product name:

1-16 characters

*Industry of Product:

Select

*Product category:

Select

Select

Select

Product profile:

1-200 characters

Technical parameters

*Networking mode:

wifi

Mobile cellular network

*Device access protocol:

MQTT (old)

To create products of other protocol suites, go to corresponding protocol suite to create

*Operation system:

Linux

Android

VxWorks

µC/OS

None

*Network operator:

CMCC

China Telecom

China Unicom

Other

Ok

Cancel

Figure 6: add device

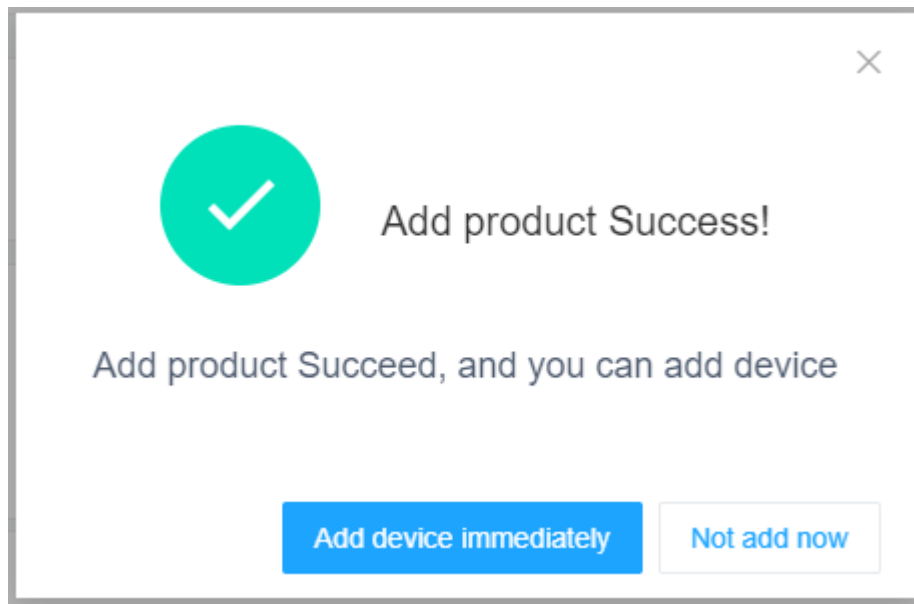


Figure 7: add production

Step 3: add equipment, enter the "equipment list" interface, click "add equipment", fill in the information to complete the equipment addition

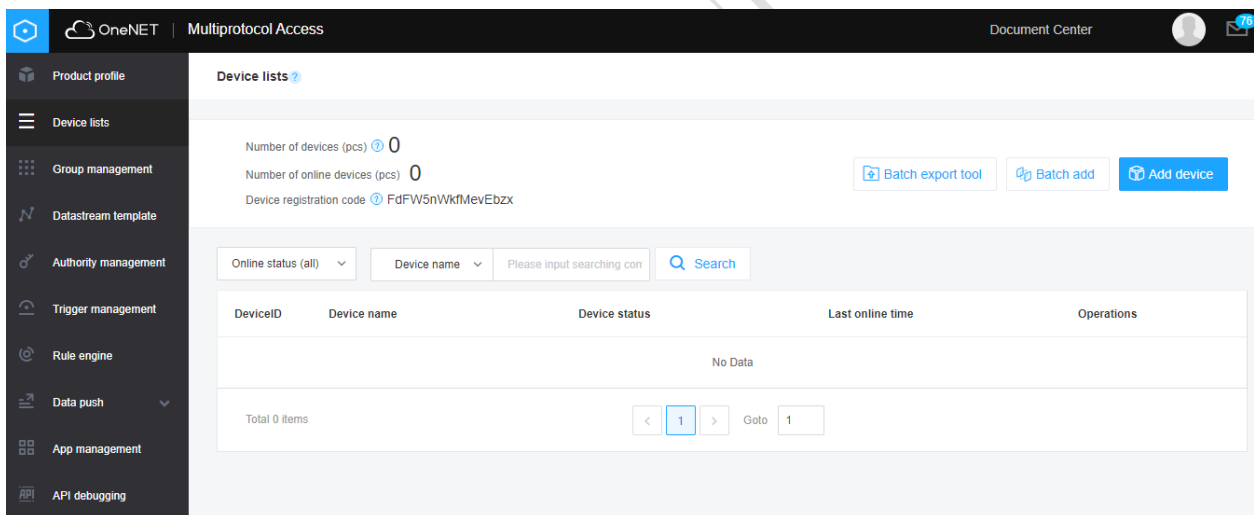


Figure 8: device lists

Add a new device
×

* Device name:

ec_gas_dev0

* Authentication information:

ecgas

* Data confidentiality:

☒ Private
☐ Open

Device description:

1-512 words

Device label:

1-8 words, up to 5 tags

Add tag

Device location:

Add

Cancel

Figure 9: add device

After the device is added successfully, you can see that the device is not online in the device list

OneNET | Multiprotocol Access

Document Center

76

Product profile

Device lists

Group management

Datastream template

Authority management

Trigger management

Rule engine

Data push

App management

API debugging

Device lists ?

Number of devices (pcs) 1

Number of online devices (pcs) 0

Device registration code FdFW5nWkfMevEbzx

Batch export tool

Batch add

Add device

Online status (all)

Device name

Please input searching con

Search

DeviceID	Device name	Device status	Last online time	Operations
604711575	ec_gas_dev0	Offline	-	Details Datastream More operations

Total 1 items

<

1

>

Goto

1

Figure 10: add device

Step 4: view product and device information, click "onenet" with cloud logo in the upper left corner to enter the developer center

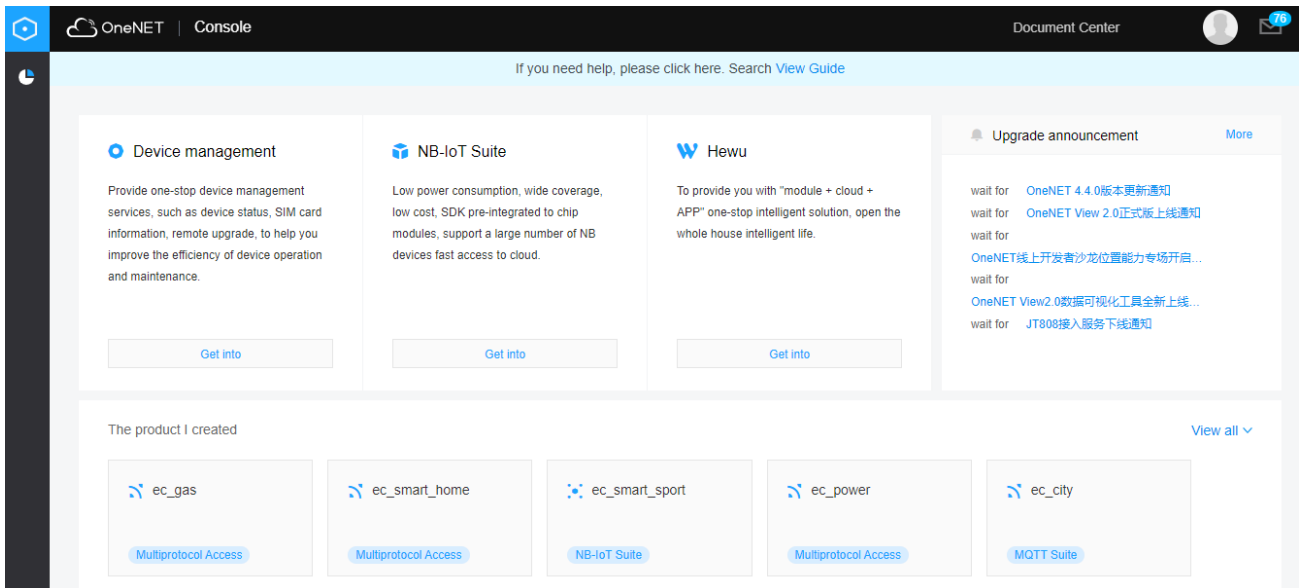


Figure 11: developer center

Click "EC power" to enter the product information interface, and remember that "product ID" is 280752. Click "equipment list" to enter, check the equipment information and remember that "equipment ID" is 55199154 and "authentication information" is ecshpower.

"Product ID", "device ID" and "authentication information" are three kinds of information necessary to create mqtt client.

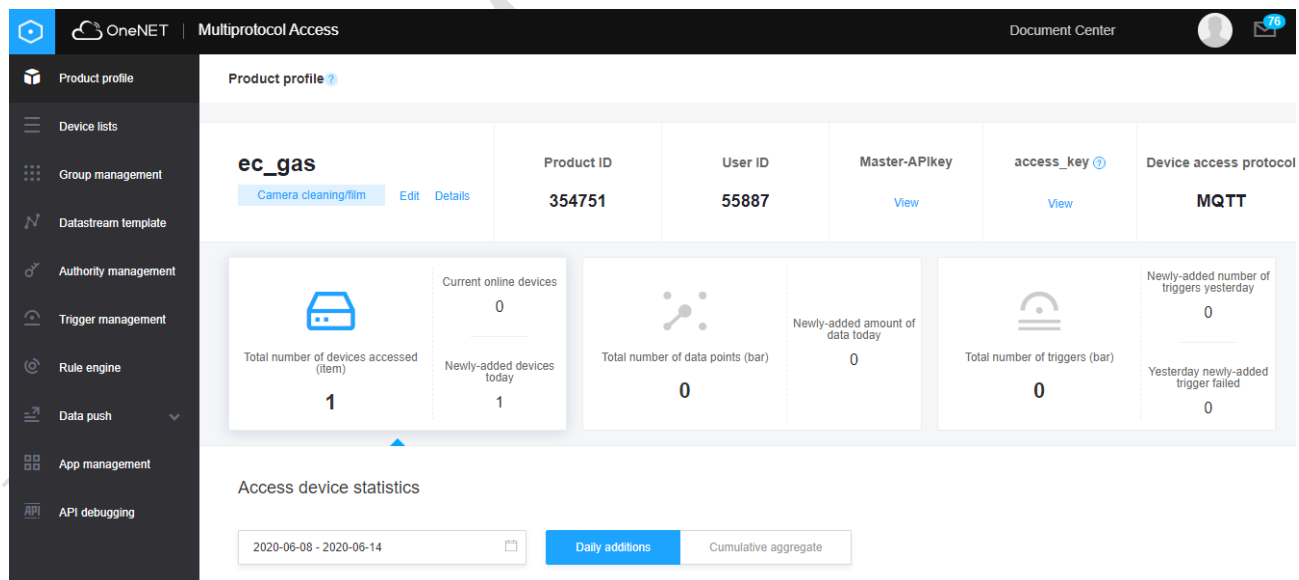


Figure 12: production information

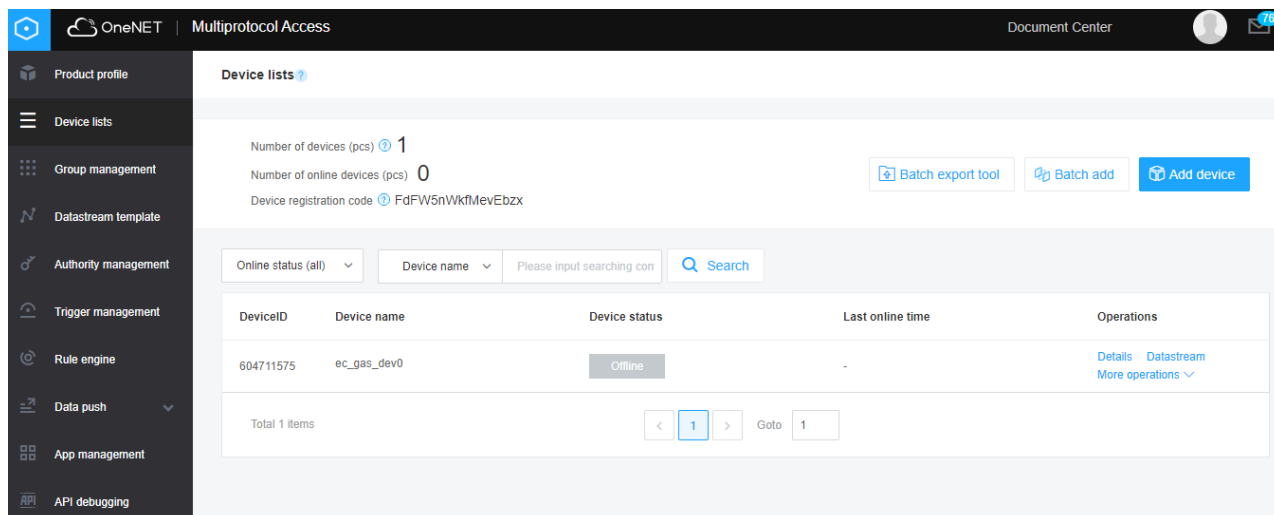


Figure 13: device information

2.2 Module side operation instructions

2.2.1 Introduction to interaction with OneNetplatform

The at command of mqtt is designed according to the functional division of mqtt, including mqtt config, mqtt open, mqtt close, mqtt connect, mqtt disconnect, mqtt sub, mqtt unsub, mqtt pub, mqtt status and mqtt receive

Mqtt config is used to configure various parameters of mqtt

The purpose of mqtt open is to create a TCP connection

Mqtt close is used to close a TCP connection

The role of mqtt connect is to connect to the onenet platform

The function of mqtt disconnect is to close mqtt connection and also close TCP connection

The role of mqtt sub is to subscribe to topic

Mqtt unsub is used to unsubscribe topics

The purpose of mqtt pub is to send data to the cloud platform

The role of mqtt status is to report the status of mqtt

The function of mqtt receive is to receive the data sent by mqtt server

The process of using mqtt's at command is as follows:

Step 1: create an mqtt connection using mqtt open and mqtt connect

Step 2: use mqtt sub to subscribe to topic

Step 3: use mqtt pub to send data

Step 4: when the mqtt server issues data, the mqtt receive command will be automatically received and printed out in the format "+ ecmtrecv: XXX"

Step 5: use mqtt disconnect to remove the mqtt connection

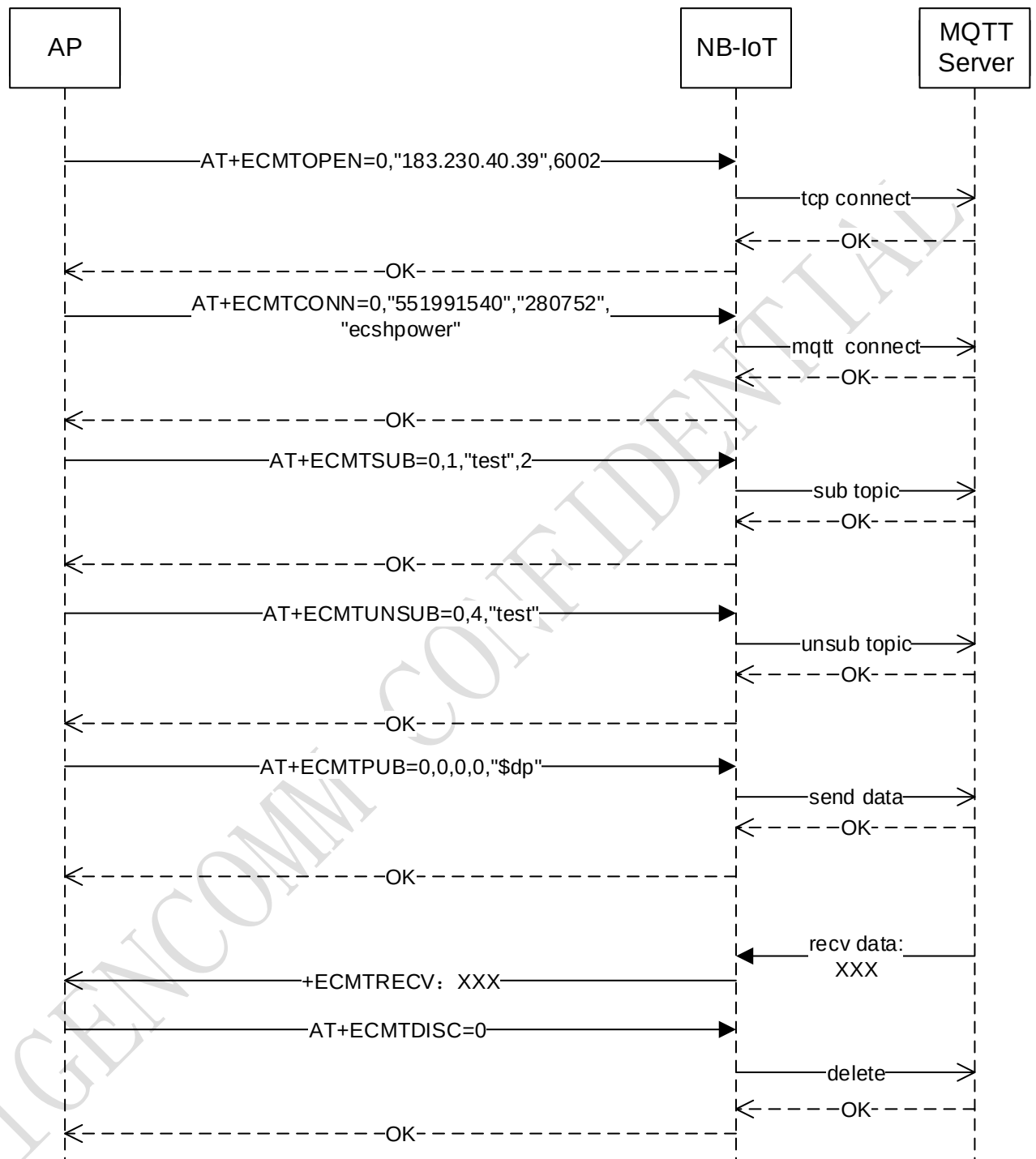


Figure 14: interaction with OneNet

2.2.2 Interact with OneNetplatform

1) Device online

Config cloud type with the command AT+ ECMTCFG ="cloud",0,1,3

When using AT+ECMTOPE=0, "183.230.40.39",6002 和 AT+ECMTCONN=0, "551991540", "280752", "ecshpower", open the "device list" of onenet, and you can see that the corresponding device status turns green, and the device displays "online"

"Product ID", "device ID" and "authentication information" are three kinds of information necessary to create mqtt client.

280752 is the "product ID", 55199154 is the "device ID", and ecshpower is the "authentication information". See Step 4 of product creation in section 2.1.2 for search examples of these three kinds of information

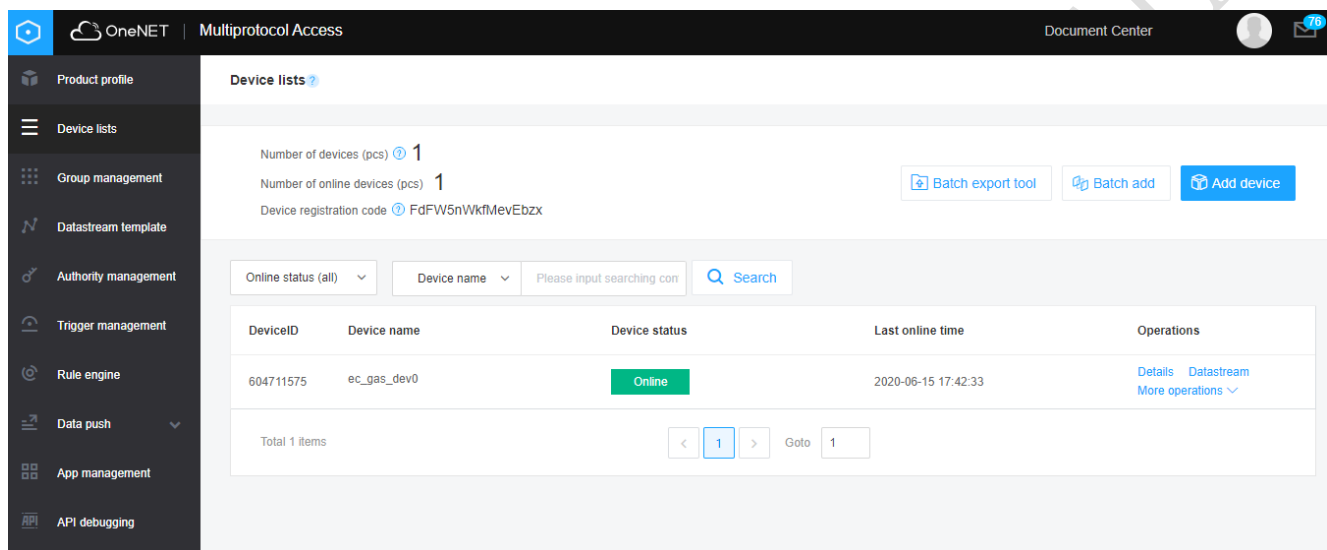


Figure 15: device status

2) Device sending data

When using AT+ECMTPUB=0,0,0,0,"\$dp" command, input 03001b7b226635f736d6172745f73656e736f725f4617461223a38387d, open the "device column" on onenet

Under "table", enter the following interface for "data flow" of online equipment. You can see that the "latest data value" is the data value just sent by the at command

The command AT+ECMTPUBsends the data according to the type 3 data format specified by onenet. The content is "{ "ec_power_data":88} ". When sending, it needs to be converted to hex format

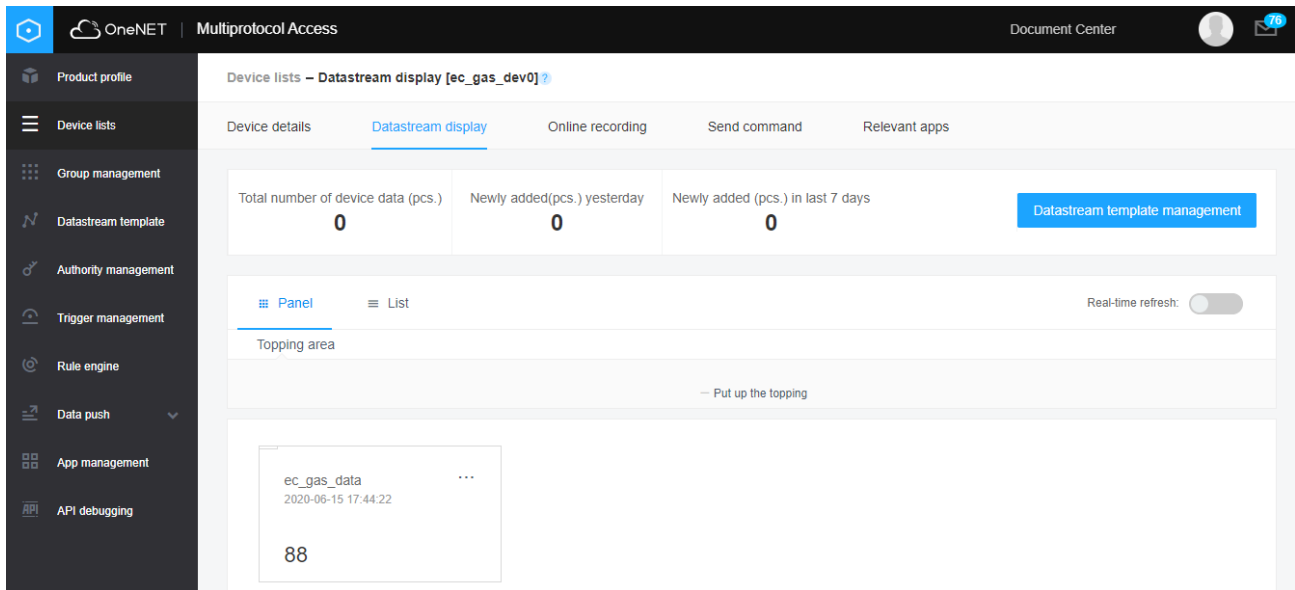


Figure 16: data flow

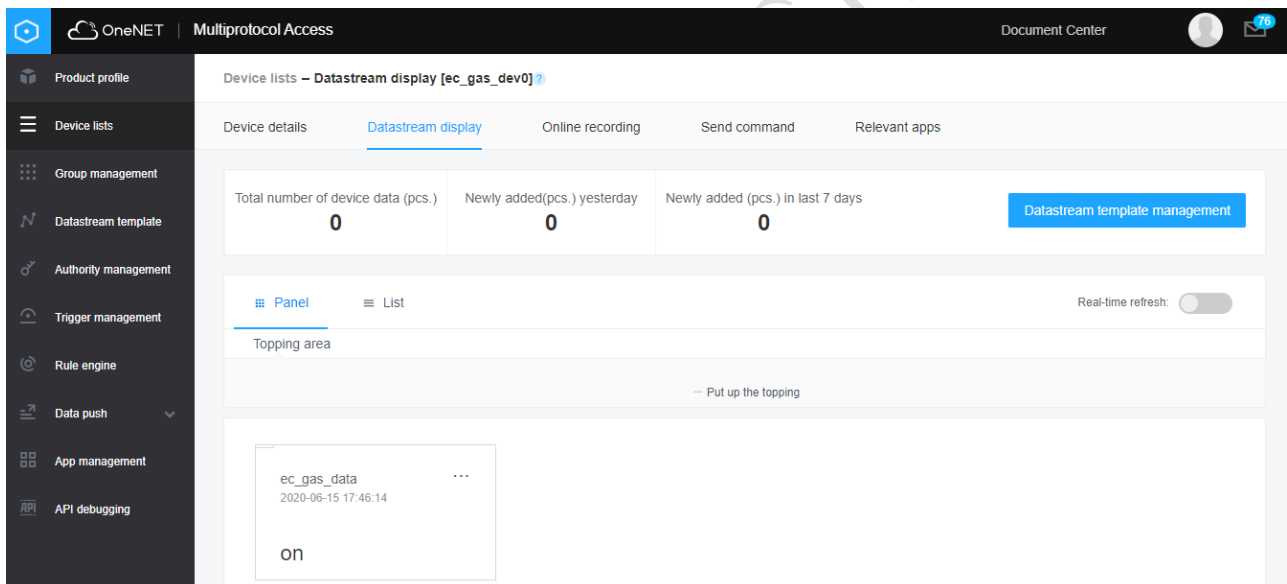


Figure 17: data flow

3) When onenet issues data, open the "device list" column on onenet, the "data flow" of online devices, and enter the following interface

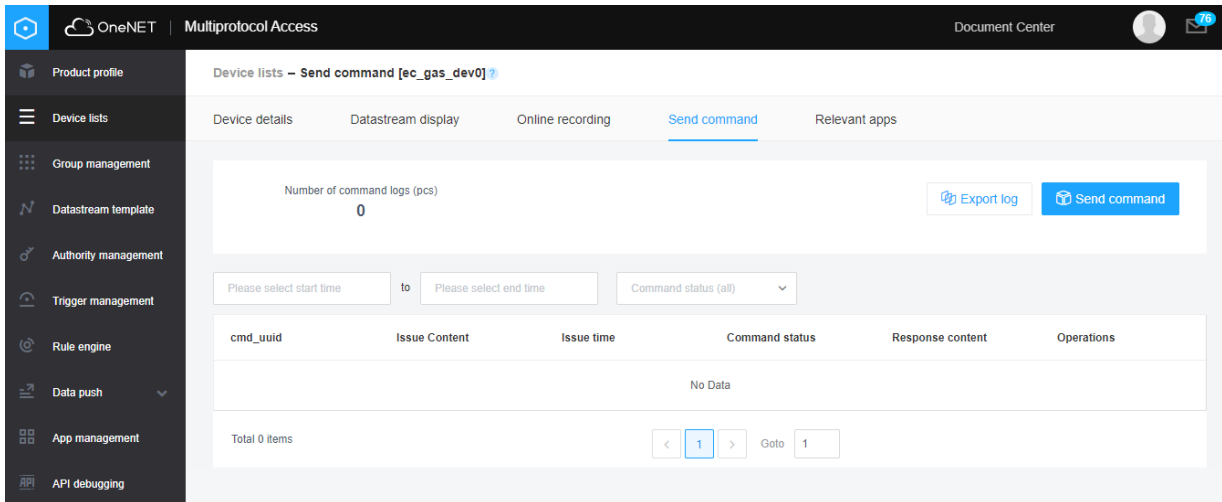


Figure 18: data flow

Click "issue command" to enter the "issue command" interface, as shown below, select "string" or "hexadecimal", input the command content, and click "send". When displaying on the serial port, there is no way to distinguish between string and hexadecimal, so the command payload received by mqtt is printed out according to string. When "hex" is selected, you may see garbled code on the serial port tool.

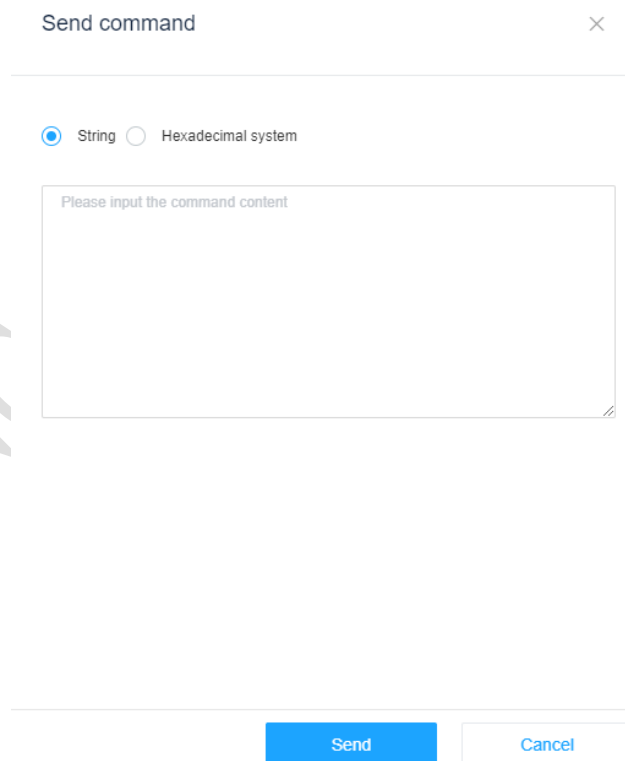


Figure 19: send data

After sending, you can see +ECMTRECV information printed out in the serial port of the device, as shown in the following figure

Note: if you choose to send hexadecimal data, UE will still print it in the form of % s after receiving it, which may result in garbled code. Printing is only for the convenience of displaying examples. The

raw data received is correct. MCU needs to add corresponding processing when receiving relevant data.

```
+ECMTRECV: 0,0,"$creq/c23e3168-87c3-5734-9bbf-30f7f389b183",eigencomm
```

Figure 20: data received form UART

3. Docking process with Alibaba Internet of things platform

3.1 Alibaba Internet of things platform side operation instructions

3.1.1 User registration

Step 1: visit

<https://help.aliyun.com/product/30520.html?spm=a2c4g.11174283.6.540.32021668tk7sOP>. Click "register",

Enter the registration page

Step 2: fill in the registration information and complete the registration

Step 3: complete real name authentication and open the Internet of things platform

Create a new Alibaba Cloud account

Get free access to 40+ products and up to 12 months usage for ECS
Enterprise-level customers can enjoy a free trial worth up to \$1300

Afghanistan

Country/region cannot be changed once registered.

Enter your email

Enter your password

Confirm your password

☐ I hereby agree to the Alibaba Cloud International Website [Membership Agreement](#), [Privacy Policy](#), [Product Terms](#) and [Terms of Use](#), under which I am contracting with Alibaba Cloud (Singapore) Private Limited.

Confirm

Figure 21: user registration

3.1.2 Product creation

Step 1: log in to Alibaba Internet of things platform

Step 2: after logging in, click "equipment management", then click "product", then click "create product", and start to create product

← Create Product (Device Model)

* Product Name

eigencomm

* Category ?

☒ Standard Category
 ☐ Custom Category

Select a standard category

View Features

* Node Type

Directly C...

Gateway s...

Gateway ...

Networking and Data Format

* Network Connection Method

Cellular (2G / 3G / 4G / 5G)

* Data Type

Custom

* Authentication Mode ?

Device Secret

^ Hide

More

Product Description

Enter a product description.

0/100

^ Hide

Save

Cancel

Figure 22: create production

Step 3: after the product is created, the following interface appears, indicating that the product has been created successfully

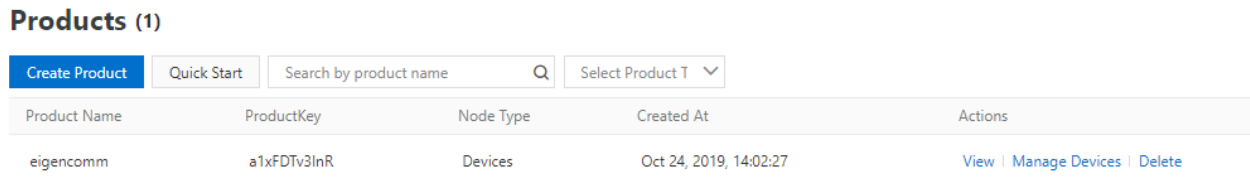


Figure 23: create production successful

Step 4: click "manage equipment" to enter the equipment management interface

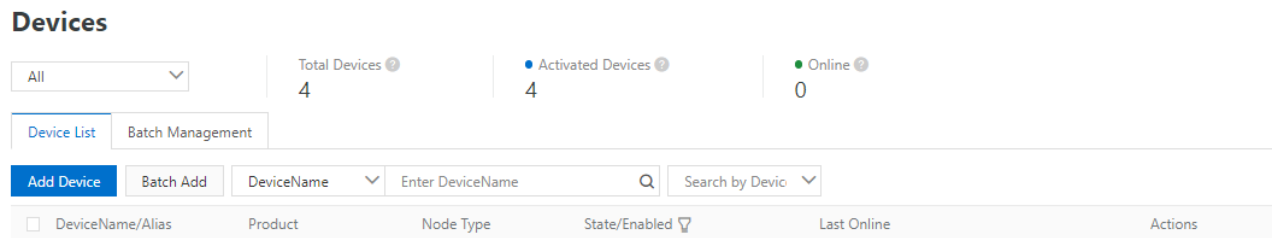


Figure 24: device management

Click "add device" and fill in devicename. After adding successfully, the device certificate will appear. Click "one click Copy" to save the device information. The productkey, devicename, devicesecret, and product name in the device certificate are the necessary information to create mqtt client.

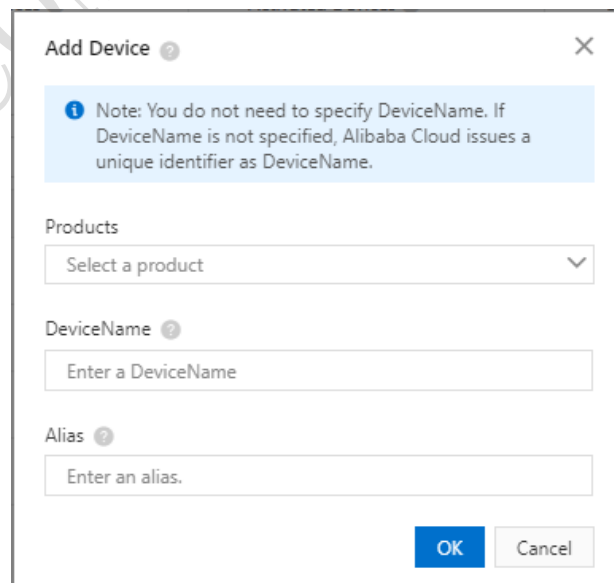


Figure 25: add device

Device Certificate

Device Certificate

Copy

ProductKey	a1xFDTv3lnR	Copy
DeviceName	ec_smoke	Copy
DeviceSecret	sWZtNMYkODMxvauyaSiGeJrVEp9jZ4Tg	Copy

Certificate Installation Modes

▼

Introduction to the unique-certificate-per-device and unique-certificate-per-product modes

Close

Figure 26: device information

After the device is created successfully, you can see that the device is not activated

Devices

All

Total Devices 4

Activated Devices 4

Online 0

Device List

Batch Management

Add Device

Batch Add

DeviceName

Enter DeviceName

Q

Search by Device

☐	DeviceName/Alias	Product	Node Type	State/Enabled	Last Online	Actions
☐	ec_pir	eigencomm	Devices	• Offlin	May 21, 2020, 16:21:53.814	View Delete
☐	ec_magnet	eigencomm	Devices	• Offlin	Oct 30, 2019, 10:55:52.530	View Delete
☐	ec_water	eigencomm	Devices	• Offlin	Jun 15, 2020, 17:01:42.267	View Delete
☐	ec_smoke	eigencomm	Devices	• Offlin	Jun 12, 2020, 14:22:02.915	View Delete

Figure 27: create device successful

3.2 Module side operation instructions

3.2.1 Introduction to interaction with Alibaba Internet of things platform

The at commands of mqtt are designed according to the functional division of mqtt, including mqtt config, mqtt open, mqtt close, mqtt connect, mqtt disconnect, mqtt sub, mqtt unsub, mqtt pub, mqtt status and mqtt receive. There are 10 at commands in total:

Mqtt config is used to configure various parameters of mqtt

The purpose of mqtt open is to create a TCP connection

Mqtt close is used to close a TCP connection

The role of mqtt connect is to connect to the Alibaba Internet of things platform

The function of mqtt disconnect is to close mqtt connection and also close TCP connection

The role of mqtt sub is to subscribe to topic

Mqtt unsub is used to unsubscribe topics

The purpose of mqtt pub is to send data to the cloud platform

The role of mqtt status is to report the status of mqtt

The function of mqtt receive is to receive the data sent by mqtt server

The process of using mqtt's at command is as follows:

Step 1: create an mqtt connection using mqtt open and mqtt connect

Step 2: use mqtt sub to subscribe to topic

Step 3: use mqtt pub to send data

Step 4: when the mqtt server issues data, the mqtt receive command will be automatically received and printed out in the format "+ ecmtrecv: XXX"

Step 5: use mqtt disconnect to remove the mqtt connection

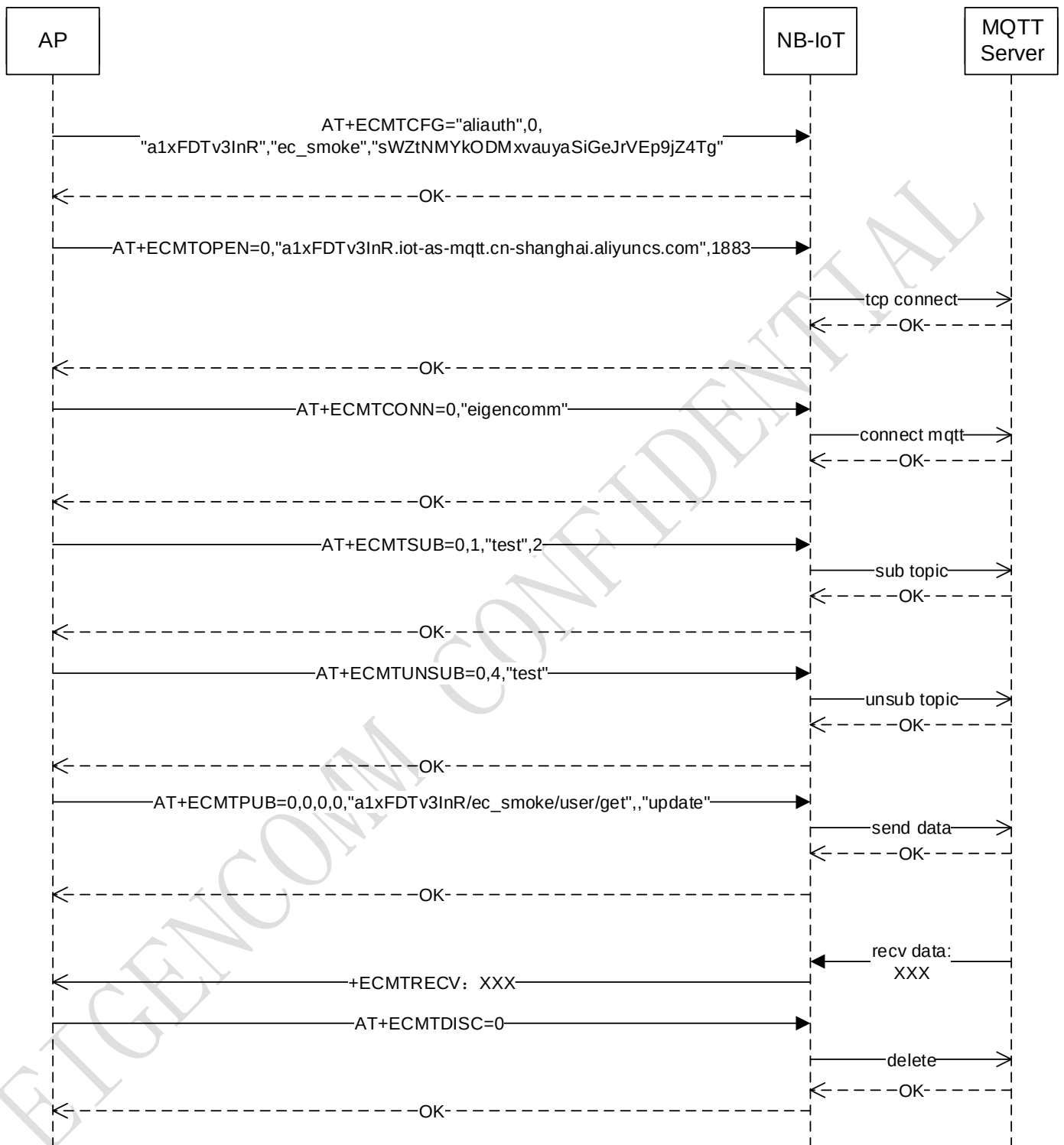


Figure 28: interaction with ali internet of things platform

3.2.2 Interactive operation with Alibaba Internet of things platform

1) Device online

Config cloud type with the command AT+ECMTCFG="cloud",0,2,1

When

AT+ECMTCFG="aliauth",0,"a1xFDTv3lnR","ec_smoke","sWZtNMYkODMxvauyaSiGeJrVEp9jZ4Tg" command is used, AT+ECMTOPE=0,"a1xFDTv3lnR.iot-as-mqtt.cn-shanghai.aliyuncs.com",1883 command and AT+ECMTCONN=0,"eigencomm",open the "device management" of the Alibaba Internet of things platform, and you can see that the corresponding device status turns green and the device displays "online"

"Product name", "devicename", "productkey" and "devicesecret" are four kinds of information necessary to create the mqtt client of Alibaba Internet of things.

In the at command, eigencomm is "product name", ec_smoke is "devicename", a1xfdtv3lnr is "productkey", sWZtNMYkODMxvauyaSiGeJrVEp9jZ4Tg is "devicesecret". See section 3.1.2 product creation for search examples of these four kinds of information

Devices

All

Total Devices 4

Activated Devices 4

Online 1

Device List

Batch Management

Add Device

Batch Add

DeviceName

Enter DeviceName

Search by Device

☐	DeviceName/Alias	Product	Node Type	State/Enabled	Last Online	Actions
☐	ec_pir	eigencomm	Devices	Offlin	May 21, 2020, 16:21:53.814	View Delete
☐	ec_magnet	eigencomm	Devices	Offlin	Oct 30, 2019, 10:55:52.530	View Delete
☐	ec_water	eigencomm	Devices	Offlin	Jun 15, 2020, 17:01:42.267	View Delete
☐	ec_smoke	eigencomm	Devices	Online	Jun 18, 2020, 20:07:50.157	View Delete

Delete

Disable

Enable

Figure 29: device online

Click "view" to see the details of the device

← **ec_smoke** Online

Products eigencomm [View](#) DeviceSecret ***** [View](#)
 ProductKey a1xFDTv3InR [Copy](#)

Device Information Topic List TSL Data Device Shadow Manage Files Device Log Online Debug

Device Information

Product Na...	eigencomm	ProductKey	a1xFDTv3InR Copy	Region	China (Shanghai)
Node Type	Devices	DeviceName	ec_smoke Copy	Authenticati...	Device Secret
Alias	Edit	IP Address	180.167.122.150	Firmware Ve...	-
Created At	Oct 24, 2019, 14:05:42	Activated At	Oct 24, 2019, 14:08:36.753	Last Online	Jun 18, 2020, 20:07:50.157
Current Stat...	Online	Real-time Delay	Test		Disabled ☐

Figure 30: device information

2) Device sending data

After using the command AT+ECMTPUB=0,0,0,0,,"a1xFDTv3InR/ec_smoke/user/get","update", under the "device details" column of the Alibaba Internet of things platform, click "log service" to enter the following interface, and then click "go to view"

← **ec_smoke** Online

Products eigencomm [View](#) DeviceSecret ***** [View](#)
 ProductKey a1xFDTv3InR [Copy](#)

Device Information Topic List TSL Data Device Shadow Manage Files **Device Log** Online Debug

Online Debug
 Device Log
 Firmware Update

Device log function has been moved to monitoring operation and maintenance
 You can also quickly jump to the device log function module by clicking the Go to View button below.

[View](#)

Figure 31: log service entry

After entering the log service, you can see that the data reported by the client has been received

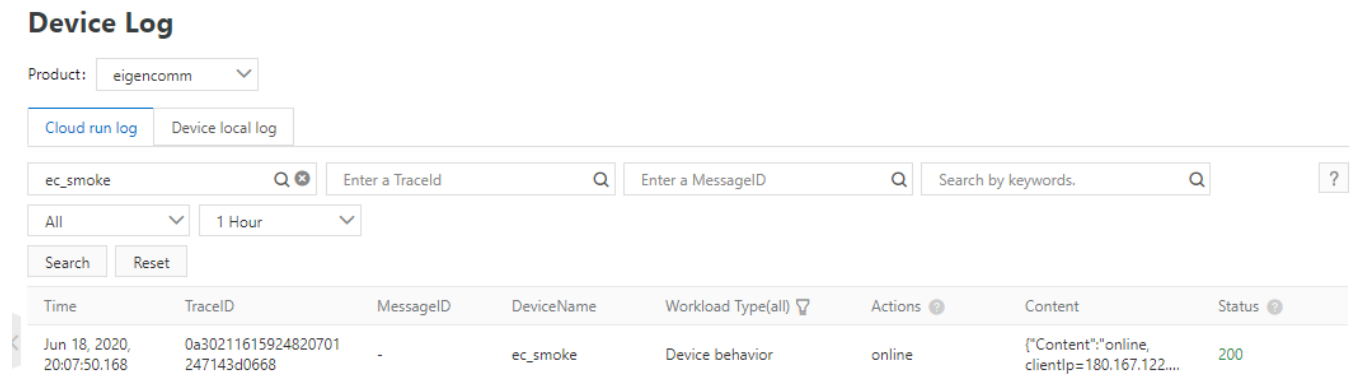


Figure 32: log service interface

3) Alibaba sending data

Alibaba Internet of things platform issues data test the data issued by Alibaba Internet of things platform needs to use the "online debugging" function of Alibaba Internet of things platform.

After the user-defined function is added, enter the device management page and select "online debugging"

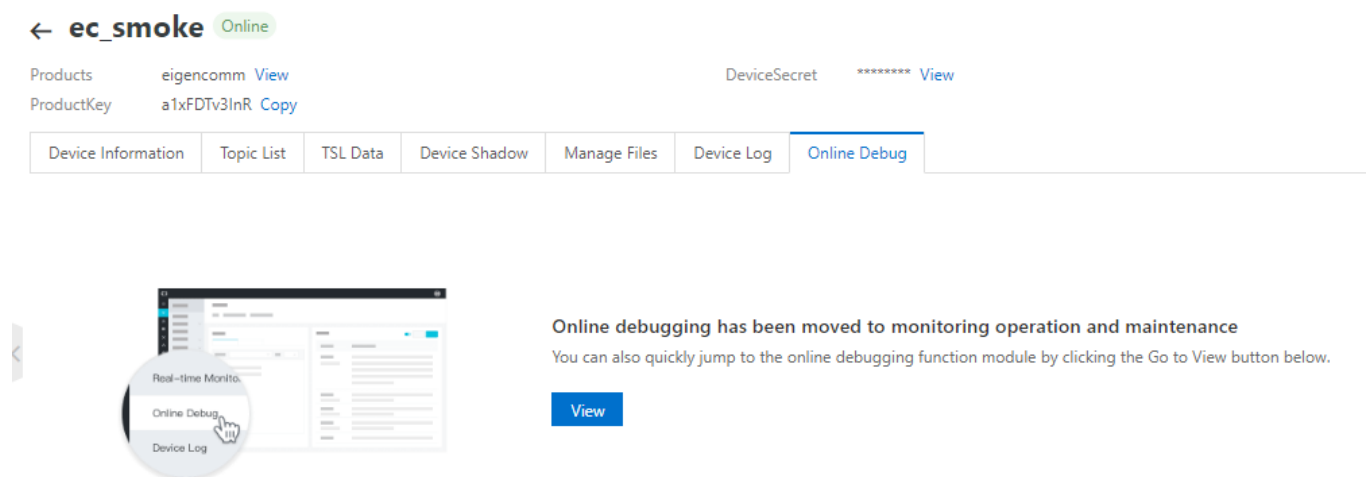


Figure 33: online debugging entrance

Click "go to view"

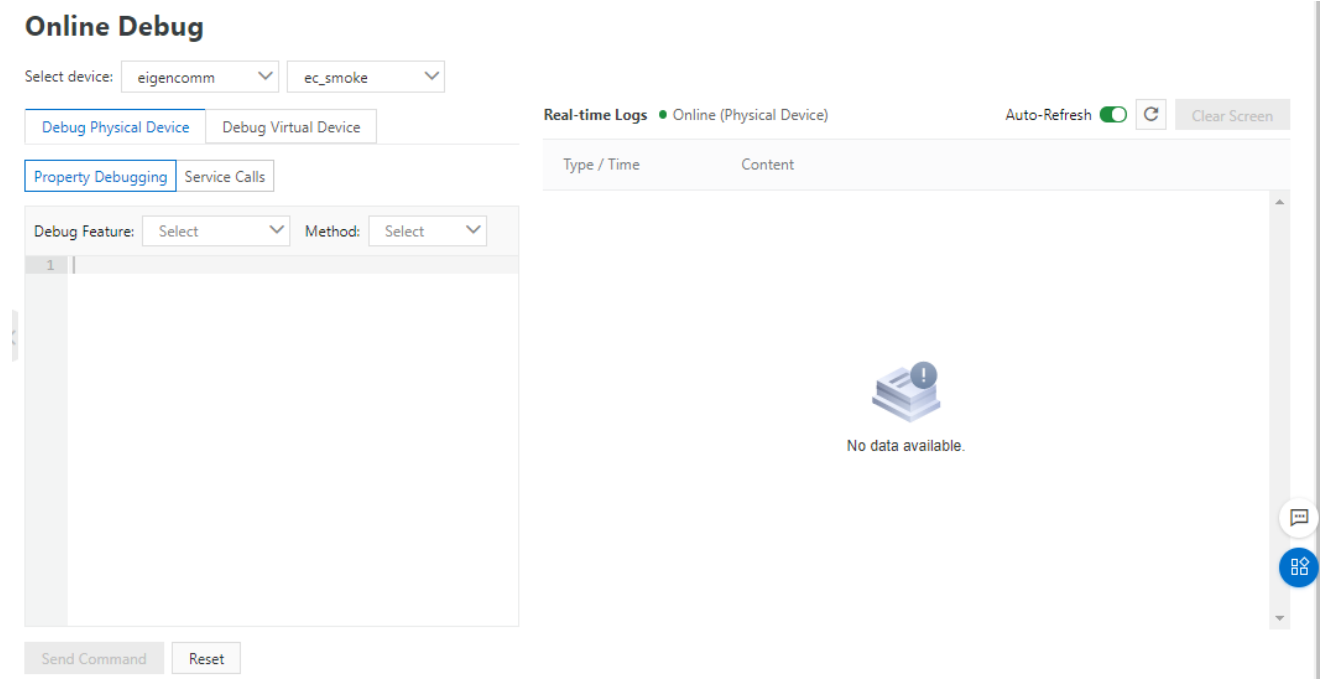


Figure 34: online debugging entrance

Select "property debugging", select "power switch" for debugging function, select "setting" for method, and then send instruction

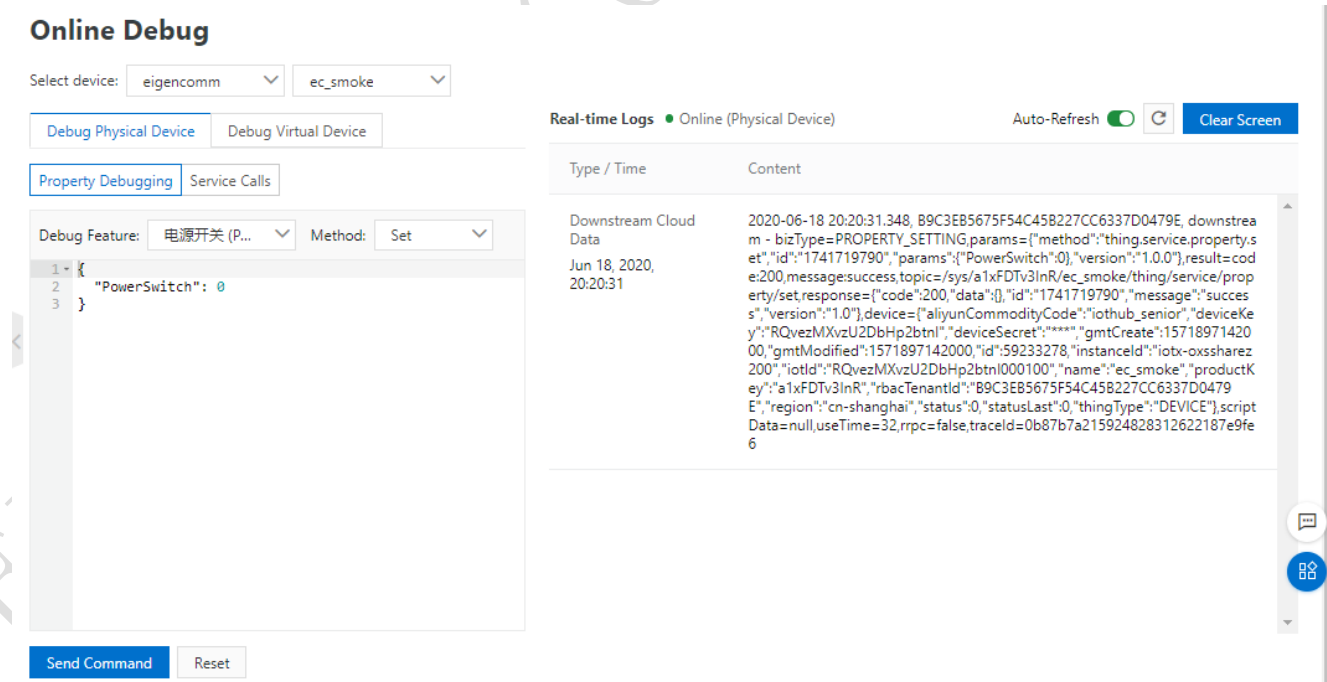


Figure 35: send message

On the serial port tool, you can see the distributed message "{" PowerSwitch ": 0}" received by the client

```
+ECMTRECV: 0,0,"/sys/alxPDTv3InR/ec_water/thing/service/property/set",
{"method":"thing.service.property.set","id":"1743710853","params":
{"PowerSwitch":0},"version":"1.0.0"}
```

Figure 36: data received form UART

4. MQTT AT Commands Details

4.1 AT+ECMTCFG

The command creates a MQTT client.

AT+ECMTCFG

Set Command Configure the keep alive time AT+ECMTCFG="keepalive",<tcpconnectID>[,<keep-alive time>]	Response OK If <keep-alive time> is omitted, query the keep-alive time: +ECMTCFG: "keepalive",<keep-alive time> OK If there is any error, response: ERROR
Set Command Configure the session type AT+ECMTCFG="session",<tcpconnectID>[,<clean_session>]	Response OK If <clean_session> is omitted, query the session type: +ECMTCFG: "session",<clean_session> OK If there is any error, response: ERROR
Set Command Configure the timeout of message delivery AT+ECMTCFG="timeout",<tcpconnectID>[,<pkt_timeout>[,<retry_times>][,<timeout_notice>]]	Response OK If <pkt_timeout>, <retry_times>, <timeout_notice> are omitted, query the timeout value of message delivery: +ECMTCFG: "timeout",<pkt_timeout>,<retry_times>,<timeout_notice>

	OK If there is any error, response: ERROR
Set Command Configure the Will information AT+ECMTCFG="will",<tcpconnectID>[,<will_fg>[,<will_qos>,<will_retain>,"<will_topic>","<will_msg>"]]	Response OK If <will_fg>, <will_qos>, <will_retain>, <will_topic> and <will_msg> are omitted, query the Will information: +ECMTCFG: "will",<will_fg>[,<will_qos>,<will_retain>,<will_topic>,<will_msg>] OK If there is any error, response: ERROR
Set Command Configure the MQTT protocol version to be used AT+ECMTCFG="version",<tcpconnectID>[,<version>]	Response OK If <version> is omitted, query the MQTT protocol version: +ECMTCFG: "version",<version> OK If there is any error, response: ERROR
Set Command Configure the device information for Ali cloud AT+ECMTCFG="aliauth",<tcpconnectID>[, "<product_key>","<device_name>","<device_secret>"]	Response OK If "<product_key>","<device_name>","<device_secret>" are omitted, query the device information: +ECMTCFG: "aliauth",<product_key>,<device_name>,<device_secret> OK If there is any error, response: ERROR
Set Command Configure the cloud type and the format of sent data for the cloud AT+ECMTCFG="cloud",<tcpconnectID>,<cloud type>,<data type>	Response OK If there is any error, response: ERROR
Test Command AT+ECMTCFG=?	Response +ECMTCFG:

	<pre>+ECMTCFG: "keepalive", (0), (0-3600) +ECMTCFG: "session", (0), (0,1) +ECMTCFG: "timeout", (0), (1-60), (1-10), (0,1) +ECMTCFG: "will", (0), (0,1), (0-2), (0,1), "will_topic", "will_msg" +ECMTCFG: "version", (0), (3,4) +ECMTCFG: "aliauth", (0), "productkey", "devicename", "devicesecret" +ECMTCFG: "cloud", (0-255), (0-255) OK</pre>
Maximum Response Time	5s
Parameter Saving Mode	NO_SAVE

Parameter

<tcpconnectID>	String type MQTT socket identifier, the value is 0
<keep-alive time>	Integer type. The range is 0-3600. The default value is 120. Unit: second. It defines the maximum time interval between messages received from a client. If the server does not receive a message from the client within 1.5 times of the keep-alive time period, it disconnects the client as if the client has sent a DISCONNECT message.
<clean_session>	Integer type. Configure the session type 0 The server must store the subscriptions of the client after it disconnects. 1 The server must discard any previously maintained information about the client and treat the connection as "clean".
<pkt_timeout>	Integer type. Timeout of the packet delivery. The range is 1-60. The default value is 10. Unit: second.
<retry_times>	Integer type. (not supported) Retry times when packet delivery times out. The range is 0-10. The default value is 3.
<timeout_notice>	Integer type. (not supported) 0 Not report timeout message when transmitting packet 1 Report timeout message when transmitting packet
<will_fg>	Integer type. Configure the Will flag 0 Ignore the Will flag configuration 1 Require the Will flag configuration

<will_qos>	Integer type. Quality of service for message delivery
	0 At most once
	1 At least once
	2 Exactly once
<will_retain>	Integer type. The Will retain flag is only used on PUBLISH messages.
	0 When a client sends a PUBLISH message to a server, the server will not hold on to the message after it has been delivered to the current subscribers
	1 When a client sends a PUBLISH message to a server, the server should hold on to the message after it has been delivered to the current subscribers
<will_topic>	Integer type.
	Will topic string, The maximum size is 255 bytes
<will_msg>	Integer type.
	The Will message defines the content of the message that is published to the will topic if the client is unexpectedly disconnected. It can be a zero-length message. , The maximum size is 255 bytes
<version>	Integer type. Version of MQTT protocol, the default is MQTT v3.1.1
	3 MQTT v3.1
	4 MQTT v3.1.1
<product_key>	Integer type.
	Product key issued by Ali Cloud, The maximum size is 32 bytes
<device_name>	Integer type.
	Device name issued by Ali Cloud, The maximum size is 32 bytes
<device_secret>	Integer type.
	Device secret key issued by Ali Cloud, The maximum size is 64 bytes
<cloud type >	Integer type
	0 mosquitto
	1 OneNet
	2 alibaba
	3-255 is reserved for customers
<data type>	Integer type, the range is 0-255
	For OneNet cloud
	1 OneNet data type1
	2 OneNet data type2
	3 OneNet data type3
	4 OneNet data type4
	5 OneNet data type5
	6 OneNet data type6
	8 Customer define data type
	For Ali Cloud
	1 Json data
	2 string data
	For other, data format insensitive

4.2 AT+ECMTOPE

The command is used to open a network for MQTT client.

AT+ECMTOPE

Set Command AT+ECMTOPE=<tcpconnectID>,"<host_name>",<port>	Response OK +ECMTOPE: <tcpconnectID>,<result> If there is any error, response: ERROR
Test Command AT+ECMTOPE=?	Response +ECMTOPE: (list of supported <tcpconnectID>s),"<host_name>",(list of supported <port>s) OK
Read Command AT+ECMTOPE?	Response [+ECMTOPE: <tcpconnectID>,"<host_name>",<port>] OK
Maximum Response Time	5s
Parameter Saving Mode	NO_SAVE

Parameter

<tcpconnectID>	Integer type MQTT socket identifier. The value is 0
<host_name>	String type The address of the server. It could be an IP address or a domain name. The maximum size is 100 bytes
<port>	Integer type The port of the server. The range is 1-65535
<result>	Integer type, Result of the command execution
	-1 Failed to open network
	0 Opened network successfully

4.3 AT+ECMTCLOSE

The command is used to close mqtt client.

AT+ECMTCLOSE

Set Command AT+ECMTCLOSE=<tcpconnectID>	Response OK +ECMTCLOSE: <tcpconnectID>,<result>
---	--

	If there is any error, response: ERROR
Test Command AT+ECMTCLOSE=?	Response +ECMTCLOSE: (list of supported <tcpconnectID>s) OK
Maximum Response Time	5s
Parameter Saving Mode	NO_SAVE

Parameter

<tcpconnectID>	Integer type MQTT socket identifier. The value is 0
<result>	Integer type, Result of the command execution -1 Failed to close mqtt 0 Close network successfully

4.4 AT+ECMTCONN

The command send MQTT connect packet.

AT+ECMTCONN

Set Command AT+ECMTCONN=<tcpconnectID>,"<clientID>"[, "<username>"[, "<password>"]]	Response OK +ECMTCONN: <tcpconnectID>,<result>[,<ret_code>] If there is any error, response: ERROR
Test Command AT+ECMTCONN=?	Response +ECMTCONN: (list of supported <tcpconnectID>s),"<clientID>"[, "<us ername>"[, "<password>"]] OK
Read Command AT+ECMTCONN?	Response [+ECMTCONN: <tcpconnectID>,<state>] OK
Maximum Response Time	5s
Parameter Saving Mode	NO_SAVE

Parameter

<tcpconnectID>	Integer type MQTT socket identifier. The value is 0
<clientID>	String type

	The client identifier, The maximum size is 48 bytes
<username>	String type
	User name of the client. It can be used for authentication, The maximum size is 48 bytes
<password>	String type
	Password corresponding to the user name of the client. It can be used for authentication, The maximum size is 96 bytes
<result>	Integer type
	0 send data successfully and received ACK from server
	1 send data successfully and received error ACK from server
	2 send data fail
<ret_code>	Integer type
	0 connect accepted
	1 connect refused – incorrect protocol version
	2 connect refused – invalid client identifier
	3 connect refused – server unavailable
	4 connect refused – bad user name or password
	5 connect refused – not authorised
	6 connect fail

4.5 AT+ECMTDISC

The command send MQTT disconnect packet.

AT+ECMTDISC

Set Command AT+ECMTDISC=<tcpconnectID>	Response OK +ECMTDISC: <tcpconnectID>,<result> If there is any error, response: ERROR
Test Command AT+ECMTDISC=?	Response +ECMTDISC: (list of supported <tcpconnectID>s) OK
Maximum Response Time	5s
Parameter Saving Mode	NO_SAVE

Parameter

<tcpconnectID>	Integer type
	MQTT socket identifier. The value is 0
<result>	Integer type, Result of the command execution
	-1 Failed to disconnect network
	0 Disconnect network successfully

4.6 AT+ECMTSUB

The command send MQTT subscribe packet.

AT+ECMTSUB

Set Command AT+ECMTSUB=<tcpconnectID>,<msgID>,"<topic>"<topic>",<qos>	Response OK +ECMTSUB: <tcpconnectID>,<msgID>,<result>[,<value>] If there is any error, response: ERROR
Test Command AT+ECMTSUB=?	Response +ECMTSUB: (list of supported <tcpconnectID>s), (list of supported <msgID>s), "<topic>", (list of supported <qos>s) OK
Maximum Response Time	5s
Parameter Saving Mode	NO_SAVE

Parameter

<tcpconnectID>	Integer type MQTT socket identifier. The value is 0
<msgID>	Integer type Message identifier of packet. The range is 1-65535
<topic>	String type Topic that the client wants to subscribe to or unsubscribe from. The maximum length is 255 bytes
<qos>	Integer type Message QoS, can be 0,1 or 2
<result>	Integer type 0 send data successfully and received ACK from server 1 send data successfully and received error ACK from server 2 send data fail
<value>	Integer type It is granted qos levels

4.7 AT+ECMTUNS

The command send MQTT unsubscribe packet.

AT+ECMTUNS

Set Command	Response
--------------------	-----------------

AT+ECMTUNS=<tcpconnectID>,<msgID>,"<topic>"	OK +ECMTUNS: <tcpconnectID>,<msgID>,<result> If there is any error, response: ERROR
Test Command AT+ECMTUNS=?	Response +ECMTUNS: (list of supported <tcpconnectID>s),(list of supported <msgID>s),"<topic>" OK
Maximum Response Time	5s
Parameter Saving Mode	NO_SAVE

Parameter

<tcpconnectID>	Integer type MQTT socket identifier. The value is 0
<msgID>	Integer type Message identifier of packet. The range is 1-65535
<topic>	String type Topic that the client wants to subscribe to or unsubscribe from. The maximum length is 255 bytes
<result>	Integer type 0 send data successfully and received ACK from server 1 send data successfully and received error ACK from server 2 send data fail

4.8 AT+ECMTPUB

The command send MQTT publish packet .

AT+ECMTPUB

Set Command AT+ECMTPUB=<tcpconnectID>,<msgID>,<qos>,<retain> ,"<topic>","<payload>"	Response OK +ECMTPUB: <tcpconnectID>,<msgID>,<result>[,<value>] If there is any error, response: ERROR
Test Command AT+ECMTPUB=?	Response +ECMTPUB: (list of supported <tcpconnectID>s),(list of supported <msgID>s),(list of supported <qos>s),(list of

	supported <retain>s), "<topic>", "<msg>" OK
Maximum Response Time	5s
Parameter Saving Mode	NO_SAVE

Parameter

<tcpconnectID>	Integer type MQTT socket identifier. The value is 0
<msgID>	Integer type Message identifier of packet. The range is 0-65535. It will be 0 only when <qos>=0
<qos>	Integer type Message QoS, can be 0,1 or 2
<retain>	Integer type 0 Server should not retain the message 1 Server should retain the message
<topic>	String type Topic that needs to be published. The maximum length is 255 bytes
<payload>	String type or Hex type Message that needs to be published. Maximum length is 1024 bytes. If in data mode, the maximum length is 1024 bytes
<result>	Integer type 0 send data successfully and received ACK from server 1 send data successfully and received error ACK from server 2 send data fail
<value>	Integer type It is granted qos levels

4.9 +ECMTSTAT

This is an unsolicited message to indicate MQTT client receive data from MQTT server.

+ECMTSTAT

+ECMTSTAT: <tcpconnectID>,<err_code>

Parameter

<tcpconnectID>	Integer type MQTT socket identifier. The value is 0
<err_code>	Integer type. Error code 1 Connection is closed or reset by peer

4.10 +ECMTRECV

This is an unsolicited message to indicate MQTT client receive data from MQTT server.

+ECMTRECV

+ECMTRECV: <tcpconnectID>,<msgID>,<topic>,<data>

Parameter

<tcpconnectID>	Integer type
	MQTT socket identifier. The value is 0
<msgID>	String type
	The message identifier of packet
<topic>	string type
	The topic that received from MQTT server
<data>	String type
	Receive data from server

5. Appendix

5.1 power on inspection process

- (1) AT //judge whether the module is powered on successfully
- (2) AT+CFUN=1 //flight mode off
- (3) AT+CEREG? //judge the attachment status of PS domain. If the second parameter is 1 or 5, the attachment is normal

5.2 Interaction process with OneNet

- (1) AT+ECPMUCFG=1,2
- (2) AT+ ECMTCFG ="cloud",0,1,3
- (3) AT+ECMTOPE=0, "183.230.40.39",6002
- (4) AT+ECMTCONN=0, " 551991540", "280752","ecshpower" //create mqtt and register the device on onenet
- (5) AT+ECMTSUB=0,1,"test",2 //subscribe to topic
- (6) AT+ECMTUNSUB=0,4, "test" //unsubscribe from topic
- (7) AT+ECMTPUB=0,0,0,0,"\$dp"
 > 03001b7b2265635f736d6172745f73656e736f725f64617461223a38387d //send data to Onenet, the content is{"ec_power_data":88}
- The command AT+ECMTPUB=0,0,0,0,"\$dp", "{"ec_power_data":88}" can send data directly
- (8) +ECMTRECV: 0,ec,topic,ec_power //receive the data ec_power from OneNet
- (9) AT+ ECMTDISC =0 //delete mqtt client and mqtt connection

5.3 Interaction process with Alibaba Internet of things

- (1) AT+ECPMUCFG=1,2
- (2) AT+ ECMTCFG ="cloud",0,2,1
- (3) AT+ECMTCFG="aliauth",0,"a1xFDTv3lnR","ec_smoke","sWZtNMYkODMxvauyaSiGeJrVEp9jZ4Tg"
//Configure Alibaba Internet of things information
- (4) AT+ECMTOPE=0, "a1xFDTv3lnR.iot-as-mqtt.cn-shanghai.aliyuncs.com",1883 //establishing tcp
- (5) AT+ECMTCONN=0, " eigencomm" //create mqtt and register the device on the Alibaba Internet of things platform
- (6) AT+ECMTSUB=0,1,"test",2 //subscribe to topic
- (7) AT+ECMTUNSUB=0,4, "test" //unsubscribe from topic
- (8) AT+ECMTPUB=0,0,0,0,"a1xFDTv3lnR/ec_smoke/user/get","update" // send data to

Alibaba Internet of things platform

(9) +ECMTRECV:

```
0,ec,/sys/a1xFDTv3lnR/ec_smoke/thing/service/property/set,{"method":"thing.service.property.set","id":
":"1879030995","params":{"PowerSwitch":0},"version":"1.0.0"} //received the data
"PowerSwitch":0
```

(10) AT+ECMTDISC=0 //delete mqtt client and mqtt connection

6. Version

Version	Date	Comments
Draft	2018-11-11	
A	2018-12-01	
V1.2	2020-02-20	

7. About US

Shanghai EigenCOMM Technology Co.,Ltd is established in Feb 2017 in Zhangjiang Hi-Tech park, Shanghai China (www.eigencomm.com) . EigenCOMM focuses on cellular based IoT chipset and solution. With accumulation of experience over the years in algorithm, L1/L2/L3 protocol, SoC, RF , platform & application software as well as the communication architecture and system, team makes its way to IoT industry.

The ultra-low power NB-IoT chipset EC616(S) has already enter mass production stage. The 4G Cat.1 chipset EC618 is in engineering samples stage, 5G is also in the R&D stage.

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